

QUBO Formulation for Quantum-Enhanced Hybrid Precoding

From the existing mmWave channel / beam / gain pipeline to QAOA

1. What we have computed now

The current QUBO code does NOT run QAOA. It converts the existing user-beam assignment problem into a Quadratic Unconstrained Binary Optimization (QUBO) model. The input is the gain matrix G , and the output is a Q matrix that encodes the assignment objective and assignment constraints.

Current configuration: $K = 20$ users, $M = 16$ candidate beams, and $N_{RF} = 8$ RF chains. One binary variable is created for every possible user-beam pair, so K times $M = 320$ binary variables and the direct QUBO matrix is 320 by 320.

2. Relationship with the existing project code

Stage	Computation	Output
1	Saleh-Valenzuela channel model	H_{users} / channel vectors
2	Oversampled DFT beam dictionary	$F = \text{beam_dict}$
3	Channel-beam gain calculation	$G[k,m] = h_k f_m ^2$
4	New QUBO module	QUBO matrix Q
5	QAOA - next step	Binary solution x^*
6	Decode	User-beam assignment
7	Evaluation	SINR, SE, EE, etc.

Your uploaded files show this separation: the beam dictionary comes from array geometry; the channel comes from the propagation model; G combines the channel and beam dictionary; Greedy uses G for its assignment; OMP uses residual channel-beam correlations; and the SINR module evaluates the final assignment with signal, interference, and noise.

3. Binary decision variable

Define one binary variable for every candidate user-beam pair:

$$x[k,m] \in \{0,1\}$$

$x[k,m] = 1$ means beam m is assigned to user k . $x[k,m] = 0$ means it is not assigned.

The implementation flattens the two-dimensional variables into a vector x of length $K \cdot M$. With $M=16$, $x[0,0]$ is variable 0, $x[0,1]$ is variable 1, and $x[1,0]$ is variable 16.

4. Input to QUBO: the gain matrix

The existing gain calculation is:

$$G[k,m] = |h_k f_m|^2$$

Here h_k is user k 's channel vector and f_m is candidate beam m . Thus G has shape K by M . For the current experiment, G is 20 by 16. A large $G[k,m]$ means beam m gives strong channel gain to user k .

5. Base assignment objective

The current QUBO optimizes the sum of selected gains:

$$\text{maximize Sum over } k,m \text{ of } G[k,m] x[k,m]$$

Since QUBO is written as a minimization:

$$E_{\text{gain}}(x) = - \text{Sum over } k, m \text{ of } G[k, m] x[k, m]$$

6. Assignment constraints

6.1 At most one beam per user

For every user k :

$$\text{Sum over } m \text{ of } x[k, m] \leq 1$$

If two beams are selected for the same user, a positive pairwise penalty is added.

6.2 At most one user per beam

For every beam m :

$$\text{Sum over } k \text{ of } x[k, m] \leq 1$$

Selecting the same beam for two different users produces a positive penalty.

6.3 Exactly N_{RF} assignments

With N_{RF} RF chains:

$$\text{Sum over } k, m \text{ of } x[k, m] = N_{\text{RF}}$$

This equality is converted into an unconstrained penalty:

$$E_{\text{count}}(x) = P_c (\text{Sum over } k, m x[k, m] - N_{\text{RF}})^2$$

For binary variables, x squared equals x . Expansion gives a linear coefficient $P_c(1 - 2 N_{\text{RF}})$ and a quadratic coefficient $2 P_c$ for each pair $i < j$. Because the implementation uses symmetric Q and $E = x^T Q x$, each off-diagonal Q entry receives P_c .

7. Complete QUBO formulation

The intended QUBO energy is:

$$E(x) = - \text{Sum } G[k, m] x[k, m] + P_u \text{Sum}_k \text{Sum}_{(m < n)} x[k, m] x[k, n] + P_b \text{Sum}_m \text{Sum}_{(k < l)} x[k, m] x[l, m] + P_c (\text{Sum } x[k, m] - N_{\text{RF}})^2$$

Then:

$$E(x) = x^T Q x, \quad x \in \{0, 1\}^{(K \times M)}$$

The optimizer therefore searches for a low-energy binary vector that prefers high-gain pairs while satisfying the user, beam, and RF-chain constraints.

8. Where SINR, delay, and other objectives fit

Important: the current QUBO code does not yet optimize SINR or delay. It optimizes the gain-based assignment and uses SINR later for evaluation. This is the correct description of the current implementation.

The current SINR model is:

$$\text{SINR}_k = P |h_k f_{mk}|^2 / (P \text{ Sum over } j \neq k |h_k f_{mj}|^2 + \sigma^2)$$

The current evaluation computes $R_k = \log_2(1 + \text{SINR}_k)$, total spectral efficiency as $\text{Sum } R_k$, and energy efficiency as total SE divided by the modeled total power.

9. Making SINR part of the QAOA objective

SINR is a ratio and its interference depends jointly on several selected beams. Therefore it cannot simply replace G in the present linear QUBO term without a derivation. A quadratic-compatible approach is to use a signal-reward term together with pairwise interference penalties.

$$\text{Signal coefficient: } S[k,m] = P |h_k f_m|^2$$

A pairwise leakage coefficient can be defined from channel-beam leakage, for example $C[(k,m),(l,n)]$ using $|h_k f_n|^2$ when both assignments are selected.

A generic quadratic surrogate is:

$$\text{maximize Sum alpha } G[k,m]x[k,m] - \text{Sum beta } C[(k,m),(l,n)]x[k,m]x[l,n]$$

After changing signs for minimization, these coefficients become entries of Q. The exact interference coupling and weights must be defined by the project's chosen objective.

10. Adding delay or QoS

Delay is not defined in the current uploaded channel/gain/SINR implementation. Therefore a numerical delay term cannot be derived from the current code alone. If a per-user or per-user-beam delay cost $d[k,m]$ is defined, a linear QUBO contribution can be:

$$E_{\text{delay}} = \gamma \text{ Sum } d[k,m] x[k,m]$$

If delay is a utility to maximize, its sign is reversed. If delay depends on interactions between selected users, it may require quadratic terms or another reformulation.

11. General multi-objective QUBO

Gain, SINR, delay, spectral efficiency and energy efficiency have different units and scales, so the metrics should be normalized before combining them. A generic normalized utility is:

$$U = w_G G_{\text{norm}} + w_R R_{\text{norm}} - w_D D_{\text{norm}} - w_I I_{\text{norm}}$$

A generic final minimization form is:

$$E(x) = -U_{\text{linear}}(x) + U_{\text{interaction}}(x) + P_{\text{user}} + P_{\text{beam}} + P_{\text{count}}$$

The weights must be explicitly selected and documented. Therefore the current project should be described as gain-based QUBO until the additional SINR/delay terms are explicitly incorporated.

12. How the Q matrix is generated

Step	Operation
1	Create one binary variable for each user-beam pair.
2	Put $-G[k,m]$ on the diagonal for the gain objective.
3	Add positive pairwise terms for two beams assigned to the same user.

4	Add positive pairwise terms for the same beam assigned to two users.
5	Expand $P_c(\sum x - N_{RF})^2$ and add its linear and quadratic coefficients.
6	Store the coefficients in symmetric Q so $E = x^T Q x$ matches the intended polynomial.

13. How QAOA uses Q

QAOA does not directly receive the words user or beam. It receives a cost Hamiltonian derived from the QUBO. Starting from $E(x)=x^T Q x$, use the binary-to-spin mapping $x_i=(1-Z_i)/2$. The resulting Hamiltonian has the Ising form:

$$H_C = \text{constant} + \sum_i h_i Z_i + \sum_{(i<j)} J_{ij} Z_i Z_j$$

QAOA alternates a cost unitary based on H_C with a mixer unitary. Classical optimization adjusts the QAOA angles. Measurement produces candidate bitstrings. A low-energy valid bitstring is decoded using `variable_map` into user-beam pairs.

For the direct current formulation, 20 users times 16 beams = 320 binary variables, so the straightforward QAOA encoding uses 320 qubits.

14. Final project flow

```
H (channel) -> F (beam dictionary) -> G (20x16 gain matrix) -> QUBO Q (320x320) -> Ising cost
Hamiltonian -> QAOA -> binary solution -> 8 selected user-beam pairs -> SINR / SE / EE
evaluation.
```

Greedy and OMP remain baselines. The same channel realization and beam dictionary should be used for fair comparison, and the resulting assignments should be evaluated with the same SINR, SE and EE definitions.

15. Key explanation for a review / interview

We are not computing a new channel or beam dictionary in the QUBO stage. We take the existing channel-derived gain matrix and convert the constrained beam-assignment problem into a binary quadratic energy function. The diagonal terms represent gain preference; quadratic terms represent assignment conflicts and the exact RF-chain count. QAOA then searches for a low-energy binary configuration. After decoding, the selected assignment is passed to the existing physical SINR model for performance evaluation.

Implementation status

The uploaded project currently computes gain first and evaluates SINR after an assignment. Therefore the present QUBO is a gain-maximization assignment QUBO with constraints. To make SINR, delay, or other QoS metrics direct QAOA objectives, their mathematical terms must be explicitly defined and incorporated into a quadratic-compatible objective.